LOW-LATENCY TARGETED BIOMECHANICS FOR PITCHING TRAINING

MARIA ARMAGANIAN, KALEN MCNAMARA,
QUINCY STEIDLE, OLIVIA VALENTI

SPONSOR: DAVID BESKY, BOSTON RED SOX

ADVISOR: PROFESSOR B. KRIS JAEGER-HELTON

CONFIDENTIALITY AGREEMENT GUIDELINES

A **non-disclosure agreement** is a legal contract that restricts the sharing of proprietary or sensitive information.

This project is subject to such an agreement to preserve the competitive advantage of the Boston Red Sox.

As such, select details have been omitted or generalized accordingly.

PROBLEM STATEMENT

Accurate, **real-time** sports training data is crucial for optimizing athlete performance and preventing injuries. Current motion-tracking systems, like PitchAI and Hawkeye, often lack accessibility and/or precision. This project aims to provide **immediate** and **affordable** pitch-by-pitch biomechanical data by integrating sensor data collection with supplemental streaming sources and visualization tools. The goal is to offer **actionable feedback** that enhances performance and promotes safer, data-driven training for athletes at all levels.

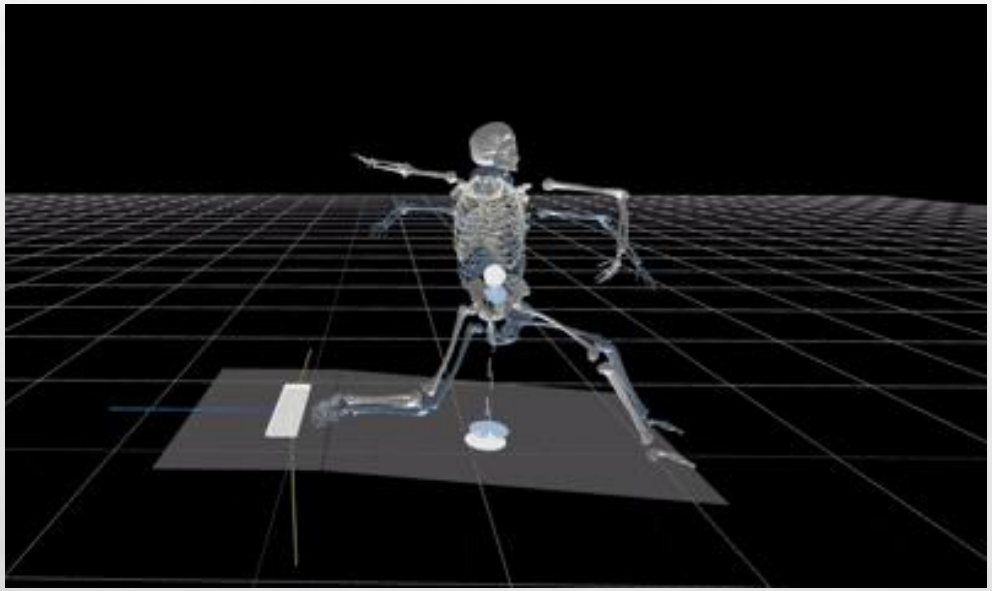
BACKGROUND

Biomechanics (n.) - *The mechanics of biological and especially musculoskeletal activity (as in locomotion or exercise); the scientific study of the action of external and internal forces on the living body.*

— Merriam-Webster Dictionary

Lead Knee Extension During Block (LKE)

- **Definition:** Change in lead knee flexion angle from foot plant to ball release (°).
- Positive values indicate knee extension (block); negative values indicate continued flexion (drift).



- **Why It Matters:**
  - Strong positive relationship with pitch velocity
  - Creates a stable front side to convert linear momentum into rotational velocity
  - Reduces energy leakage and improves kinetic chain efficiency

BACKGROUND

BACKGROUND

WHY REAL-TIME BIOMECHANICS *WAS* INACCESSIBLE



Cost & Infrastructure Barriers

High accuracy requires expensive hardware ecosystems.

Multi-camera, high-speed motion capture arrays (8–20+ synchronized cameras)



Calibration & Setup Complexity

Precise multi-camera calibration and controlled environment protocols

Standardized marker placement procedures



Accuracy Tradeoffs in Field-Ready Systems

Lower sampling frequency systems reduce biomechanical fidelity.

2D markerless capture lacks depth accuracy and suffers from occlusion



Time Burden (Collection + Processing)

Marker placement and preparation are time-intensive

Data cleaning and third-party processing can take hours to days

TEAM APPROACH: DISTRIBUTED EXPERTISE MODEL

Intentional Team Formation:

- Assembled based on complementary strengths:
  - Mechanical engineering paired with industrial
  - Sports domain knowledge paired with coding expertise.

Parallel Workstream Structure

- Tasks were owned by the team member with the most relevant expertise
- Ensures depth and accountability across all project components

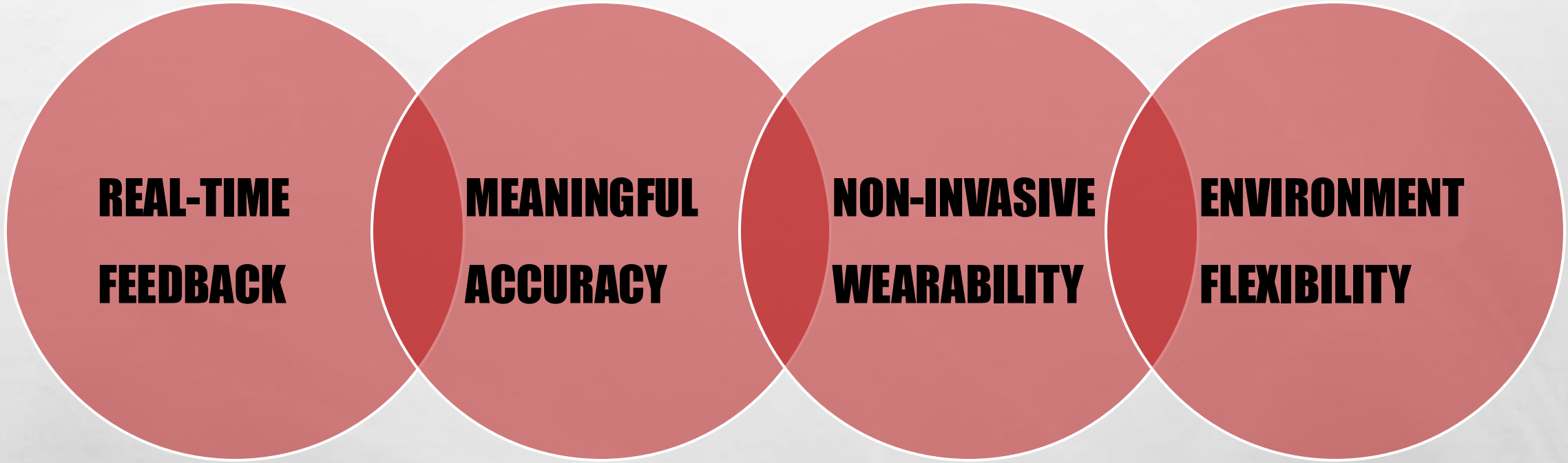
Planning Sessions

- Group meetings were reserved for high-level planning and scoping rather than redundant co-working time.

Collaborative Tooling

- Active branching strategy in GitHub enabled asynchronous collaboration
  - Develop, review, and merge work independently while maintaining version control and traceability.

CRITICAL SUCCESS FACTORS

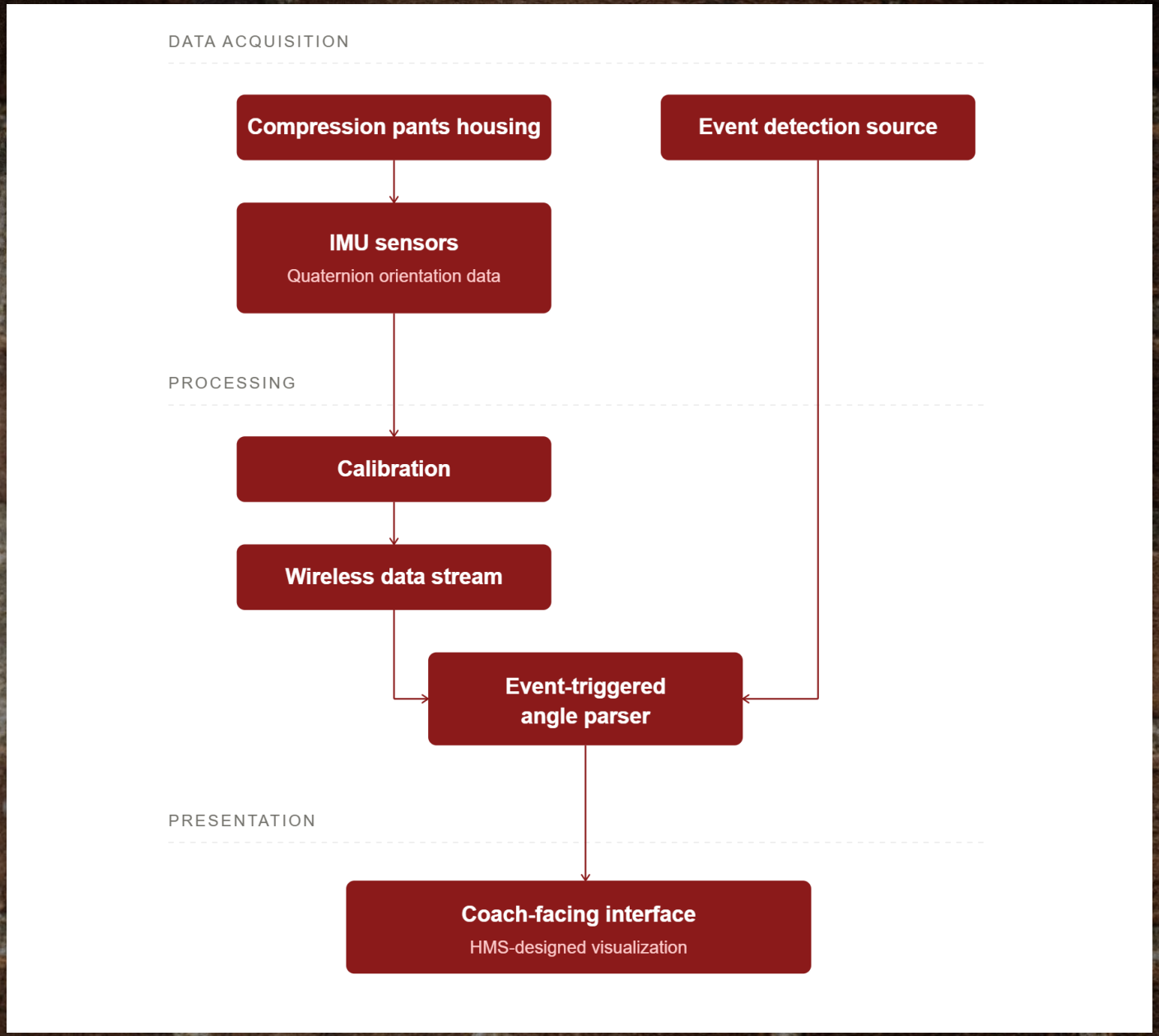


These Critical Success Factors define the conditions required for our system to be usable in real training environments. We evaluate each CSF using measurable KPIs to ensure technical performance aligns with usability

INDUSTRIAL ENGINEERING TOOLS APPLIED

- **SYSTEMS ENGINEERING:** MULTI-COMPONENT INTEGRATION
- **HUMAN FACTORS:** CALIBRATION TIME, VISUALIZATION CLARITY, INTERFACE DESIGN
- **QUALITY ENGINEERING**: ROOT MEAN SQUARE ERROR, BLAND-ALTMAN, INTRACLASS CORRELATION COEFFICIENT
- **PROCESS OPTIMIZATION:** REDUCING LATENCY TO UNDER 1 MINUTE
- **RISK MANAGEMENT:** INCREMENTAL TESTING STRATEGY
- **WORK MEASUREMENTS/TIME STUDY:** STRUCTURED TESTING INTERVALS USING TIME EFFICIENCY
- **EXPERIMENTAL DESIGN:** ENVIRONMENTAL CHANGES FOR FIELD TESTING

OVERALL SYSTEM ARCHITECTURE



IMU SENSOR PIPELINE

BUILT TO SUPPORT RELIABLE REAL-TIME CAPTURE, INTERPRETATION, AND EXPORT

QSense IMUs

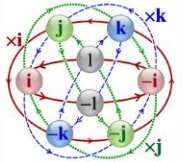
Wireless **inertial measurement units (IMUs)** provide **real-time** motion orientation data



Qsense IMU



Wireless Dongle



$$q = w + xi + yj + zk$$

Sensor orientation is represented as **quaternions** a four-parameter rotation format for **3D** tracking

Pipeline Flow



IMU DATA STREAM

CALIBRATION





TELEMETRY CAPTURE

PARSER



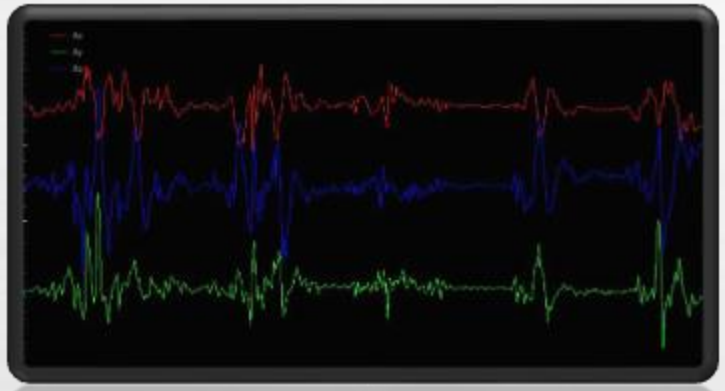


TARGETED METRICS

EXPORT



DEVELOPMENT & TESTING

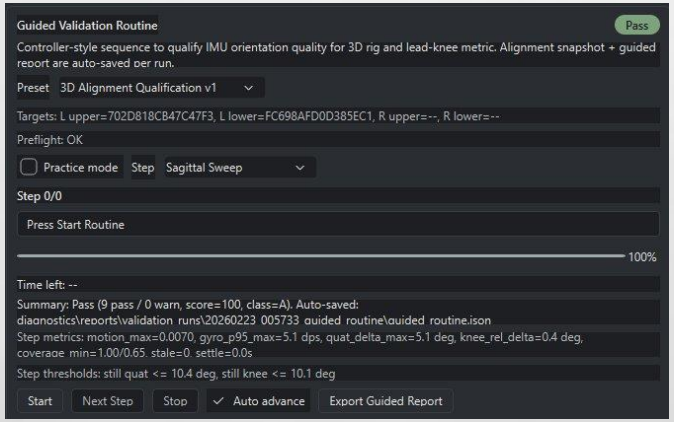


System Capabilities

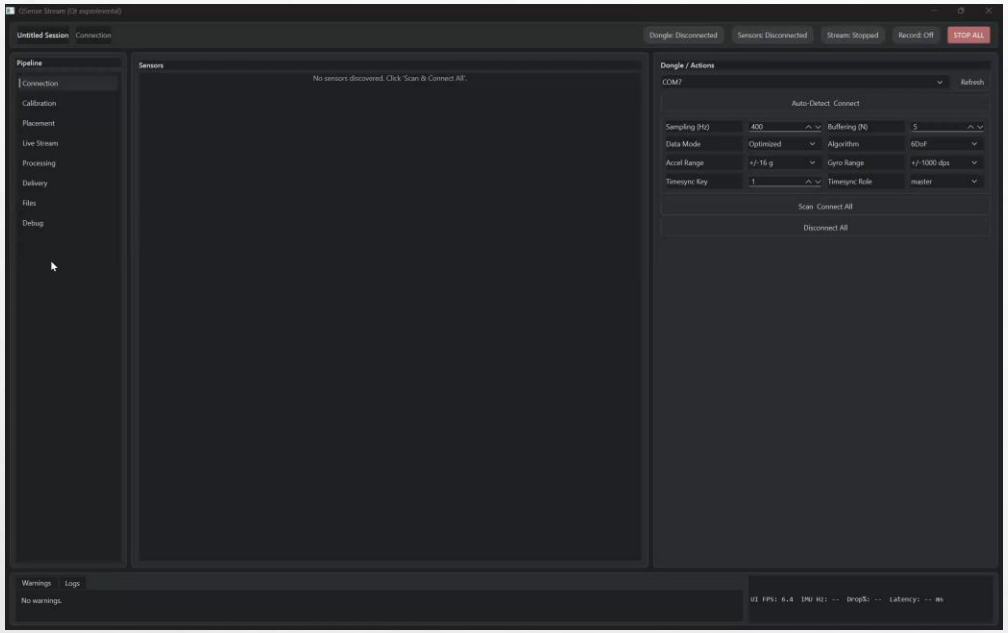
- High-rate quaternion streaming (multi-IMU)
- Time synchronized sensor pipeline
- Automated validation & QA checks
- End-to-end pipeline (capture → metrics → export)

Lead Knee Validation

- Calibration routine
- Reliable joint interpretation
- Repeatable motion results







Initial IMU Pipeline Prototype Interface

Key Challenges Addressed

- Stream instability → stabilized high-rate data
- Sensor sync issues → aligned multi-sensor timing
- Data reliability → automated QA + preflight checks

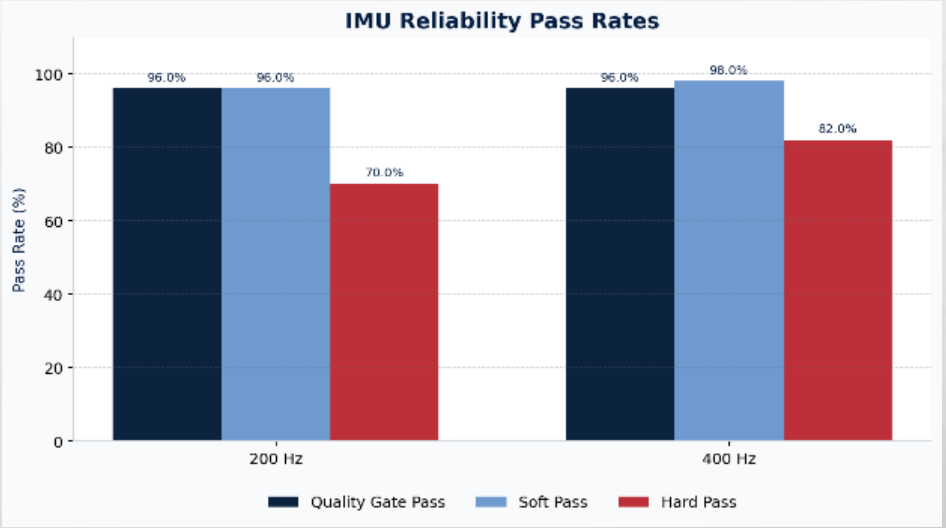
Physical Verification & Validation

Verification (System Reliability)

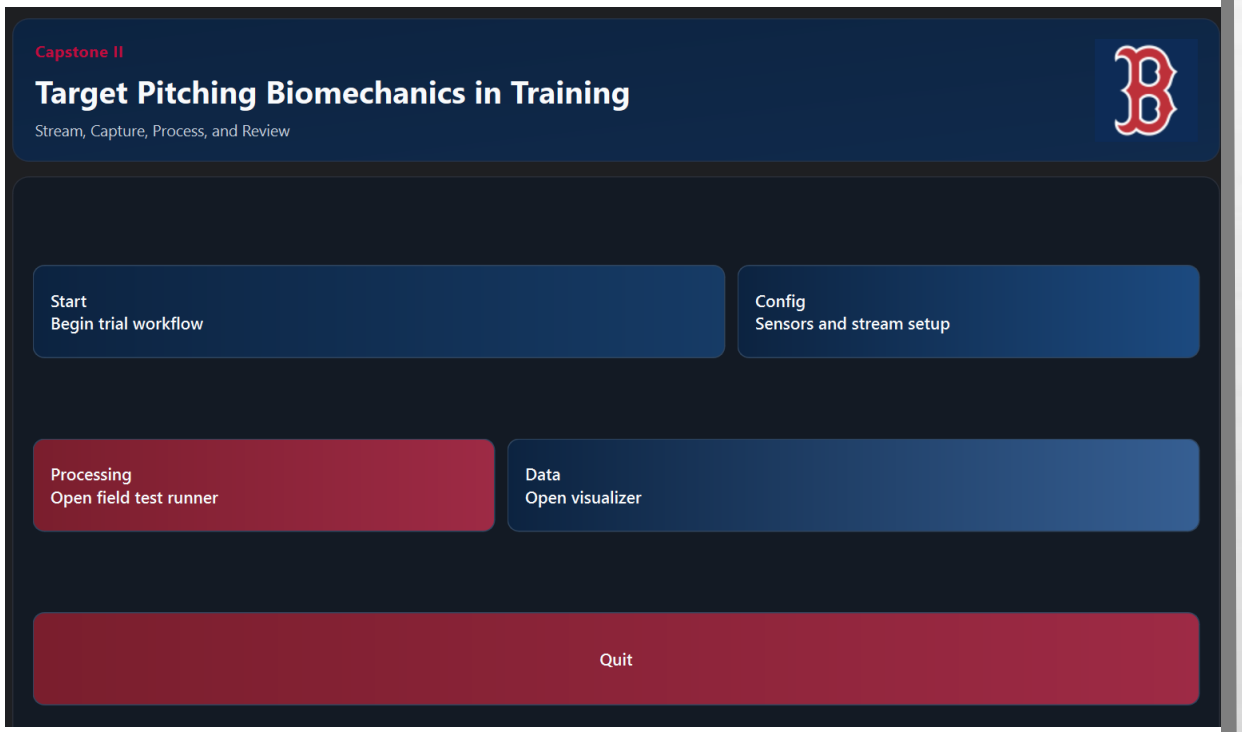
- Stream integrity (timing + continuity)
- IMU reliability across sessions
- Consistent quaternion → kinematic outputs

Validation (Real-World Accuracy)

- Controlled motion trials (raise, extend, rotate)
- Cross-validation with reference capture







PIPELINE ADJUSTMENTS

DEVELOPED A FINALIZED TEST INTERFACE FOR IMU CONFIGURATION, DATA COLLECTION, PROCESSING, EXPORT AND VISUALIZATION

INTERFACE DESIGN

HMS Principles

- **High Data Density**

  FP angle, BR angle, extension delta, thigh/shank breakdown, timing, and orientation all accessible in a single view.

- **Diagnostic Transparency**

  Raw time-domain plot preserves actual event timing, enabling experts to verify foot plant and ball release detection accuracy.

- **Power-User Tooling**

  Toggleable overlays (Source Channels, Knee Diagnostics, Key Events) give users granular control over what's visualized.

- **Spatial Grounding**

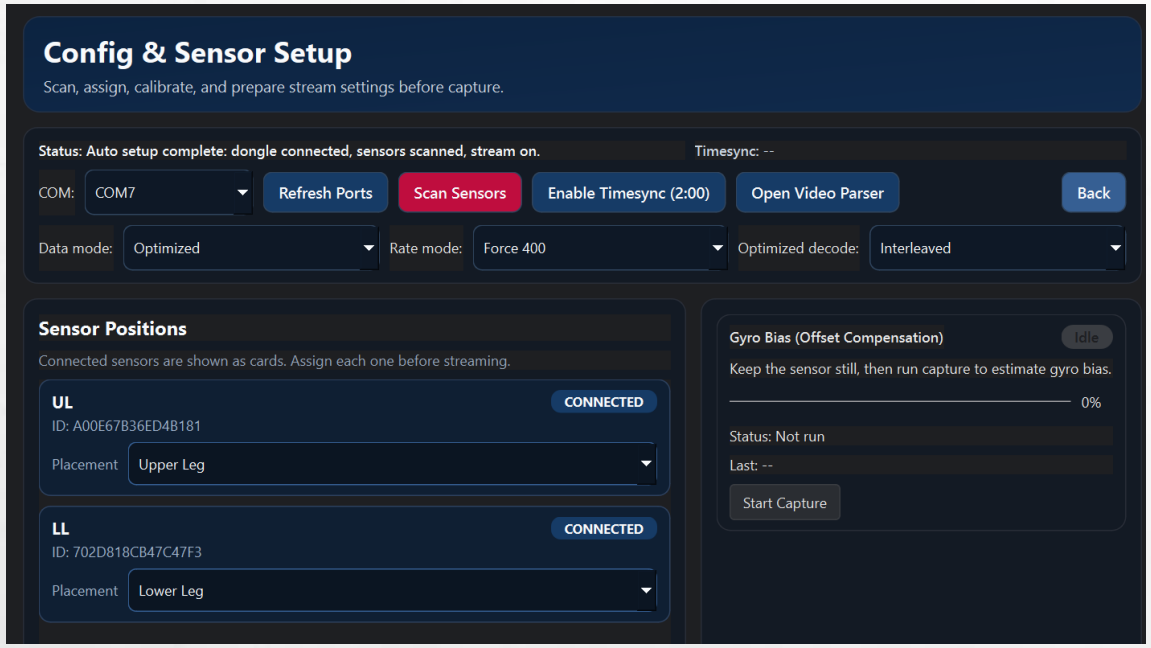
  Knee orientation stick-figure diagram anchors abstract degree values in physical body context, bridging numbers to movement.

- **Field-Ready Palette**

  Dark background with high-contrast text supports readability in variable lighting conditions.

- **Component Decomposition**

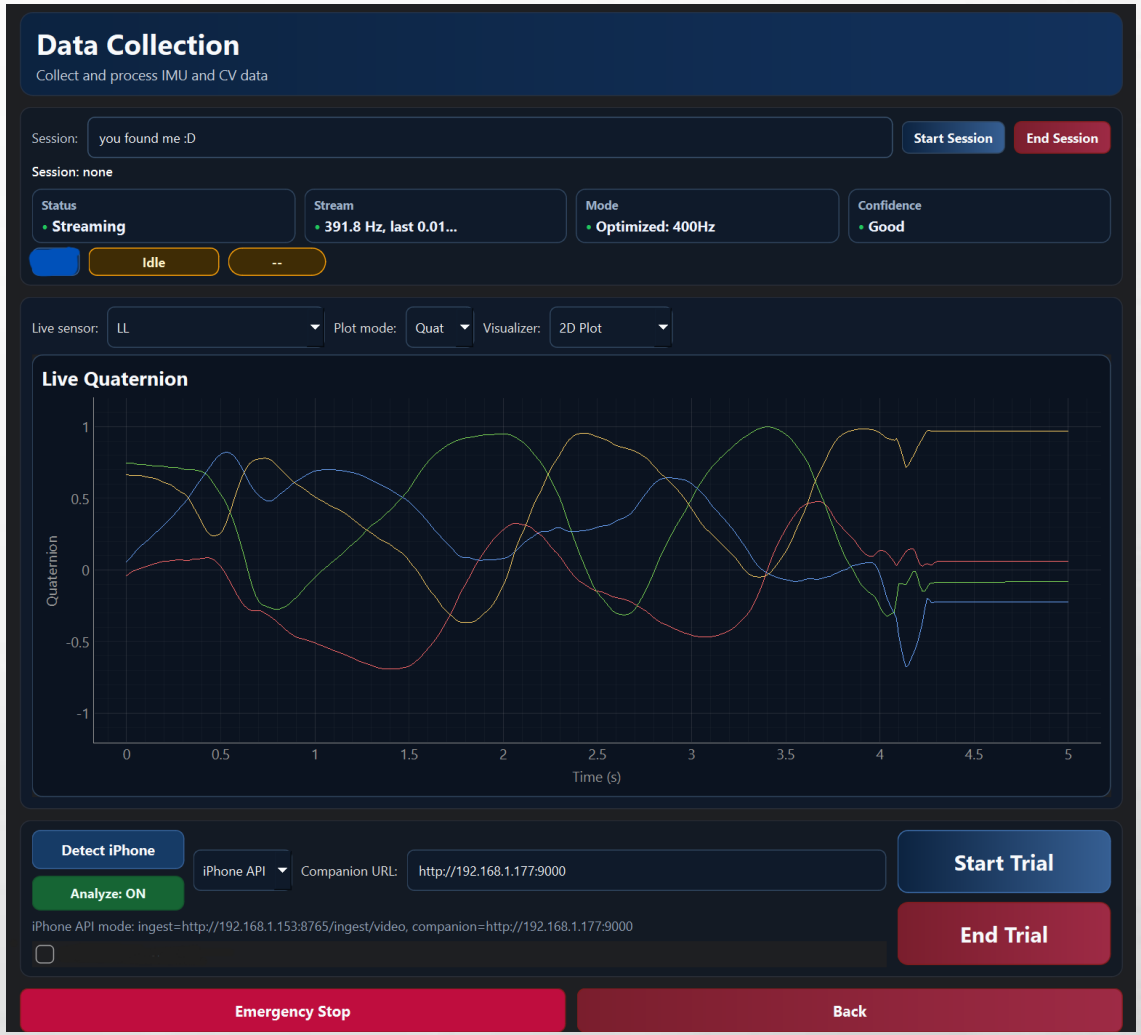
  Separate display of thigh and shank angles at FP and BR allows users to diagnose which segment is driving the extension pattern.



IMU Configuration & Setup Screen

Final Testing Interface

- Unified workflow for setup, capture, and validation
- Real-time visualization of multi-IMU motion data
- Integrated tools for calibration, QA, and trial control
- Designed for repeatable, field-ready data collection



Pitch Trial Data Collection & Stream Visualization



Data Visualization & Targeted Metric



Kinematic + Event-Based Metric Derivation

WEARABLE SYSTEM DESIGN

DESIGN OBJECTIVE

Develop a wearable mounting system that supports consistent IMU anatomical alignment during high-velocity pitching motion without restricting athlete mobility.

BASE GARMENT SELECTION

Men's lightweight compression leggings:

- Provide uniform circumferential compression
- Reduce soft tissue artifacts (STA)
- Minimize relative sensor displacement
- Allow full lower-body range of motion during pitching



WEARABLE SYSTEM DESIGN

Design Inspiration

Adapted from elite wearable telemetry systems (catapult tracking apparel)

- Integrated pocket-based device mounting
- Maintains orientation during explosive movement
- Field-validated for high performance sport use





Integrated Sensor Pocket Design

- Fixed anatomical sensor placement
  - Distal femur
  - Proximal tibia
- Form-fitted IMU pocket
- Elastic compression fit
- Upper material seal



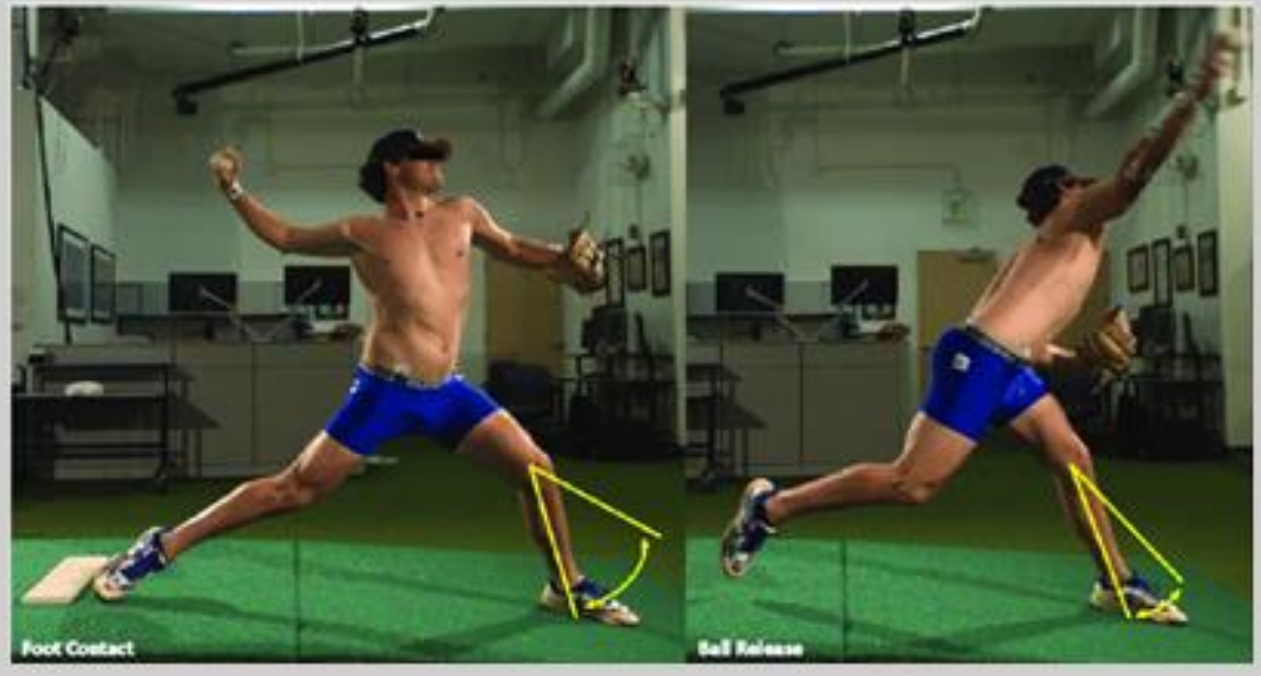
EVENT DETECTION RESULTS

92/95 complete detections (96.8%)

- Missed events for 3 trials due to data quality issues

Foot Plant (n=92)

Ball Release (n=92)



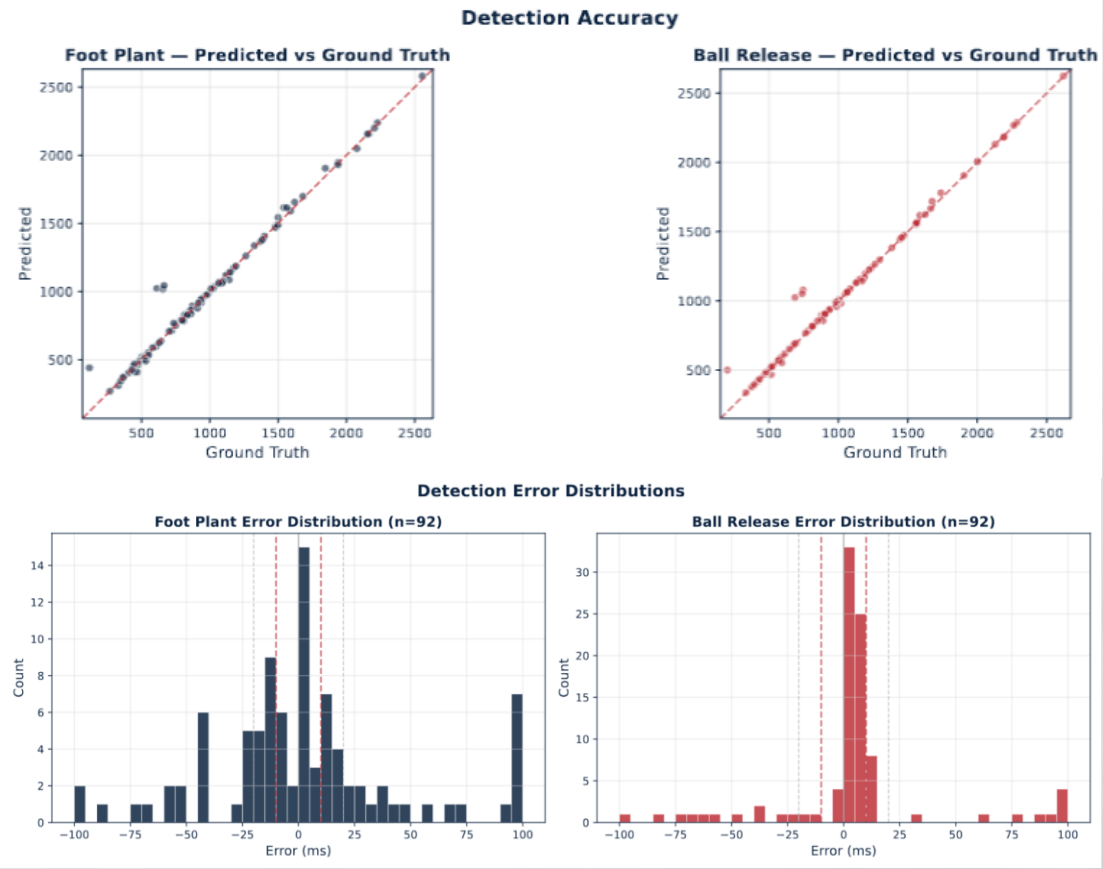
EVENT DETECTION RESULTS

Foot Plant (n=92)

- Mean/Median Absolute Error: 60.4 ms / 16.7 ms
- Within ±10 ms: 28.3%
- Within ±20 ms: 55.4%

Ball Release (n=92)

- Mean/ Median Absolute Error: 44.0 ms / 6.2 ms
- Within ±10 ms: 67.4%
- Within ±20 ms: 78.3%



KEY SOLUTIONS

Team Problem-Solving Highlights

01 Direct Vendor Collaboration

When sensor documentation proved incomplete, the team established a direct technical line with QSense, resolving configuration gaps and accelerating integration.

02 Disciplined Task Management

A heavy deliverable cadence demanded deliberate delegation and scheduling tradeoffs, keeping written work and development progress on track simultaneously.

03 Custom Test Environment

With no mound access and Boston winter conditions working against us, the team built a purpose-made mound setup and adapted protocols to test through the constraints.

04 Multi-Environment Validation

Intermittent burst errors in sensor data prompted testing across varied environments, strengthening the team's understanding of failure modes and improving data confidence.

05 Custom iOS Data Collection App

The need for supplemental biomechanical data beyond the primary sensor suite led to the development of a dedicated iPhone application for synchronized capture.

06 Multi-Stage Quality Gating

Sensor inconsistencies drove the design of a rigorous filtering and validation pipeline, ensuring only high-confidence trials informed final analysis.

PROJECT TIMELINE

September

CAPSTONE I MILESTONES

- Fixed sensor stream issues
- Validated sensor performance
- Refined scope to single-metric focus
- Finalized testing interface
- Launched field data collection

January

CAPSTONE II MILESTONES

- Wearability/usability system design
- Integrated collaborative interface
- Conducted field study in varying locations
- Data collection: broad demographics of pitchers
- Fixed major pipeline errors

April & Beyond

NEXT STEPS

- Gold standard validation with Kinatrax
- Compression sleeve production
- Continued environmental testing
  - Focus on outdoor setting
  - Interference source testing
- Extrapolation to other metrics
  - Back knee flexion at foot plant
  - Est. peak torso rotational velocity
  - Center of mass peak speed

CONCLUSION



THANK YOU!

QUESTIONS, IDEAS?

